

CHINA LAKE, CA, USA

WMO: 746120

Lat: 35.688N

Lon: 117.693W

Elev: 2230

StdP: 13.55

Time Zone: -8.00 (NAP)

Period: 07-19

WBAN: 93104

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	23.1	26.5	-7.6	4.0	47.2	-2.9	5.1	47.3	31.3	55.3	27.8	55.0	2.9	200	0.519

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	31.4	108.3	65.2	106.0	64.5	103.8	63.6	69.0	97.6	67.7	97.6	66.3	97.0	10.7	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.6	82.8	79.3	55.9	72.3	83.3	52.5	63.7	81.5	34.3	97.4	33.1	97.4	32.0	97.4	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.5	24.4	20.7	DB	16.8	112.2	3.5	1.8	14.3	113.5	12.3	114.6	10.4	115.6	7.9	117.0	(4)
(5)			WB	13.4	71.7	3.3	2.2	11.0	73.3	9.1	74.5	7.2	75.7	4.8	77.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.7	45.9	49.8	57.6	64.1	71.6	82.0	88.2	86.4	78.6	65.5	53.7	44.2	(6)	
	DBStd	16.26	5.93	6.23	6.47	6.79	7.05	6.27	4.45	4.10	6.40	6.23	6.49	5.97	(7)	
	HDD50	474	152	75	13	2	0	0	0	0	0	1	35	196	(8)	
	HDD65	2431	593	426	239	95	24	0	0	0	3	66	339	646	(9)	
	CDD50	6210	24	69	249	424	669	961	1184	1127	857	483	147	16	(10)	
	CDD65	2692	0	0	11	68	228	511	719	662	410	82	1	0	(11)	
	CDH74	43226	5	32	384	1248	3534	8109	11702	10434	6051	1551	176	0	(12)	
(13)	CDH80	26697	1	0	85	458	1838	5212	7933	6915	3639	597	19	0	(13)	
(14)	Wind	WSAvg	7.6	5.3	7.3	8.8	9.8	9.3	8.7	8.0	7.9	7.3	6.9	5.6	5.8	(14)
(15) Precipitation	PrecAvg	4.0	0.8	1.1	0.5	0.1	0.1	0.0	0.1	0.1	0.2	0.2	0.3	0.6	(15)	
	PrecMax	9.4	5.1	4.4	2.6	1.4	0.8	0.2	0.7	0.6	2.1	1.2	1.3	3.0	(16)	
	PrecMin	0.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)	
	PrecStd	2.4	1.1	1.1	0.6	0.3	0.2	0.0	0.1	0.1	0.4	0.3	0.3	0.7	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.7	77.9	87.1	93.5	101.2	110.4	111.8	108.8	104.8	93.8	82.6	69.0	(19)	
		MCWB	50.1	51.6	55.0	58.2	60.6	64.4	66.1	65.5	65.2	59.9	53.7	50.0	(20)	
	2%	DB	67.4	74.5	82.2	89.0	96.9	106.5	109.3	106.7	101.6	90.0	79.3	65.1	(21)	
		MCWB	47.1	50.5	53.6	55.8	59.0	63.4	65.6	65.1	63.1	58.9	51.9	47.4	(22)	
	5%	DB	64.0	70.5	78.1	85.4	94.0	103.9	106.8	104.8	99.2	86.6	75.2	61.8	(23)	
		MCWB	46.1	49.1	51.9	54.6	58.0	62.1	65.0	64.3	62.4	56.4	51.3	45.9	(24)	
	10%	DB	60.5	66.3	74.6	81.8	90.3	100.7	104.3	102.6	96.7	82.8	70.2	58.3	(25)	
		MCWB	45.0	47.1	50.7	53.5	57.2	61.3	64.3	63.3	61.2	54.8	48.9	44.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.5	54.7	56.5	59.5	61.8	67.4	71.0	70.7	68.6	64.3	55.4	53.5	(27)	
		MCDB	62.6	66.6	76.9	87.4	97.1	104.1	92.2	97.8	95.0	82.9	76.5	62.5	(28)	
	2%	WB	50.9	51.4	54.9	57.3	60.1	65.5	69.2	68.9	67.0	61.4	53.6	51.1	(29)	
		MCDB	59.5	70.7	75.9	85.1	92.9	100.6	98.0	97.2	95.6	81.8	74.5	60.8	(30)	
	5%	WB	49.0	50.0	53.3	55.8	59.0	63.9	68.2	67.1	65.1	59.0	52.0	48.6	(31)	
		MCDB	58.4	68.1	75.6	81.9	90.5	98.7	97.7	96.9	92.3	81.0	71.8	57.5	(32)	
	10%	WB	46.8	48.3	51.7	54.3	57.7	62.4	66.9	65.6	63.0	56.9	50.3	45.9	(33)	
		MCDB	58.5	64.3	72.4	79.1	87.9	96.7	97.4	96.8	91.4	78.3	68.5	55.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	28.0	28.8	29.3	29.9	30.7	33.1	31.4	32.7	32.7	32.1	31.2	26.6	(35)	
		MCDBR	33.7	35.8	35.9	35.1	35.9	36.6	33.9	34.1	34.8	36.7	36.7	32.0	(36)	
	5% WB	MCWBR	18.6	17.5	15.8	14.2	13.0	11.9	11.0	11.6	12.7	16.3	18.6	18.5	(37)	
		MCWBR	24.7	32.3	32.4	32.2	34.6	34.4	27.5	29.3	29.8	29.5	32.9	24.1	(38)	
(40)	Clear-Sky Solar Irradiance	taub	0.267	0.277	0.285	0.310	0.327	0.322	0.345	0.334	0.314	0.292	0.269	0.268	(40)	
(41)		taud	2.582	2.570	2.539	2.463	2.440	2.460	2.451	2.455	2.516	2.571	2.604	2.594	(41)	
(42)		Ebn at Noon	302	310	314	309	303	303	295	297	300	300	299	295	(42)	
(43)		Edh at Noon	24	27	30	35	36	36	36	35	31	27	23	22	(43)	
(44)	All-Sky Solar Radiation	RadAvg	961	1277	1751	2193	2491	2713	2523	2357	2012	1506	1106	871	(44)	
(45)		RadStd	72	117	75	81	101	96	106	69	52	77	46	65	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (28 neighbors)	+1.15	N/A	-3.06	+0.77	N/A	N/A	N/A	-209	+382	+207	(47)

Nomenclature: See separate page